

Photonic Metamaterial-Inspired Polarization Manipulating Devices on Etchless Thin Film Lithium Niobate Platform

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Photonic metamaterials interact with light at the micro–nano scale, enabling unprecedented optical manipulation capabilities, which play a key role in nonlinear optics, spin optics, negative index, and zero index materials. Recently, thin-film lithium niobate on insulator (LNOI) has emerged as a promising platform for integrated photonics due to its unique material properties, including an excellent electro–optic effect, wide transparency window, and low waveguide losses. Metamaterial structures are promising for constructing novel integrated photonic circuit building blocks on LNOI as they can provide advantageous circuit functionality. In this work, as a proof of concept, high-performance polarization manipulating devices including polarization splitter rotator (PSR), polarization rotator (PR), and polarizer are demonstrated using photonic metamaterials on the etchless LNOI platform by introducing a silicon nitride layer on the top of LNOI wafer as the loading material. In this way, full advantage of lithium niobate (LN) can be taken to achieve various high-performance integrated photonic devices, while avoiding the etching of LN and simplifying the fabrication process greatly, which can bring bright prospects for achieving large-scale lithium niobate integrated photonic circuits.

1. Introduction

The term “metamaterial” was initially coined by Prof. Rodger M. Walser to describe composites that do not exist in nature, are artificially manufactured, 3D, and have periodic structures. Metamaterials can be designed with micron/nano-scale patterns or structures that interact with waves in ways that are not possible with natural materials. For example, metamaterials can effectively manipulate electromagnetic waves by altering the basic unit structure, shape, arrangement, and other physical properties of the material. Among them, photonic metamaterials show great optical manipulation capabilities that conventional photonic structures cannot achieve.^[1] For example, it can enhance the collective nonlinear phenomenon to a level of several orders of magnitude larger than that of natural nonlinear materials,^[2] provide a route for spin-controlled nanophotonic

applications,^[3,4] realize imaging without intrinsic limitations of resolution,^[5] and compensate for the phase delay caused by anisotropic materials.^[6] By altering the duty cycle of the photonic metamaterials, it is possible to manipulate the effective refractive index of the waveguide, providing more options for achieving polarization handling devices. In other words, photonic metamaterials can be utilized to achieve dispersion manipulation to meet the desired requirements that other materials do not possess, which is conducive to the realization of wavelength-insensitive (broadband) on-chip polarization manipulating devices.

Thin-film lithium niobate on insulator (LNOI) provides a promising photonic integrated circuit platform for future optical interconnect technologies due to its attractive properties, such as strong modal confinement, low optical propagation loss, high-speed electro–optical modulation, and efficient optical nonlinearity.^[7] Lithium niobate is an anisotropic material due to the asymmetry of its crystal structure. It encounters two different refractive indices as light travels along the Y-/Z- direction in the devices on X-cut lithium niobate crystals.^[8] Therefore, the devices designed on the lithium niobate (LN) platform need to consider the influence of the propagation direction. The proposed devices in this work are designed along the

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Table 1. Comparison of simulation and experiment results of our proposed devices.

Function	Length [μm]	ER [dB]	IL [dB]	Bandwidth [nm]
PSR	160	>29 (Sim)>20 (Exp)	<1.15 (Sim)<1.83 (Exp)	85 (1515–1600) (Sim)85 (1515–1600) (Exp)
PR	500	>15 (Sim)>14 (Exp)	<0.82 (Sim)<1.07 (Exp)	160 (1515–1675) (Sim)85 (1515–1600) (Exp)
TE-pass polarizer	70	>21 (Sim)>20 (Exp)	<1.05 (Sim)<1.75 (Exp)	800 (1200–2000) (Sim)85 (1515–1600) (Exp)

IL, Insertion loss; ER, Extinction ratio

Y-direction to take advantage of the mode hybridization phenomenon. With the advancement of lithium niobate fabrication technology, various photonic devices used for optical interconnection have been proposed and demonstrated on the LNOI platform, such as modulators,^[9–13] wavelength division multiplexers,^[14–17] and mode division multiplexers.^[18–20] However, there are still challenges to achieving microstructures, especially for photonic metamaterials, on the LNOI platform by direct etching LN due to its slanted sidewall waveguides. To realize large-scale integration on LNOI and overcome the challenges of direct etching, another promising choice is etchless LNOI by introducing a loaded material on the top of the thin film LNOI and only etching the loading material to form hybrid waveguides. Currently, various materials have been investigated as loaded materials for this etchless scheme,^[21–23] and among them, silicon nitride (Si_3N_4) is one of the most promising loaded materials for lithium niobate, as it has a similar but slightly lower refractive index, a similar transparency window to lithium niobate and the availability of mature fabrication processes. The silicon nitride-loaded LNOI waveguide overcomes the waveguide sidewall angle challenges of the direct-etching LN while maintaining the excellent material properties of LNOI, which makes it suitable for fabricating photonic integrated circuit components that use photonic metamaterials.

On-chip polarization manipulation technology plays a key role in improving the performance of optical interconnection, which can double the data transmission capacity,^[24] provide phase matching conditions for nonlinear wavelength conversion,^[25,26] and provide a solution for designing polarization-independent integrated optics.^[27] In this work, as a proof of concept for photonic metamaterials with extraordinary optical manipulation capabilities, we design and fabricate photonic metamaterial structures to achieve several polarization manipulation devices including polarization splitter rotator (PSR), polarization rotator (PR), and polarizer on the silicon nitride loaded LNOI platform by using mature microfabrication process to only etch the silicon nitride layer. **Table 1** shows the comparison of the simulation and experimental results of our proposed polarization manipulation devices, from which we can see the experimental results agree well with the simulation results, except for the bandwidths of the devices due to the testing limitation of our equipment. This demonstration shows the possibility of using photonic metamaterials to construct integrated photonic devices on etchless LNOI platform, from which we can see photonic metamaterials exhibit powerful mode-field manipulation and refractive index manipulation capabilities, offering options for future high-performance LNOI integrated photonic circuits. This work also opens a bright window for the applications of photonic metamaterials on nonlinear optics, spin optics, and large-scale photonic integration on the LNOI platform in the future.

2. Principle and Design

2.1. Operation Principle

The phenomenon of birefringence is common for anisotropic optical waveguides, resulting in different propagation properties for TE- and TM-polarized light. Here, we design and fabricate artificial photonic metamaterials to achieve anisotropic waveguides to construct high-performance polarization manipulation devices on the etchless LNOI platform, in order to show photonic metamaterials have great potential for manipulating the distributions of the optical mode field, optical waveguide's refractive index, and effectively controlling of the light's polarization. The profile of the silicon nitride and lithium niobate hybrid waveguide used in this work is shown in **Figure 1d**. A 300 nm thick silicon nitride thin film is deposited and patterned on a commercially available LNOI wafer, which has a lithium niobate thickness of 300 nm to form the hybrid waveguide. In the hybrid waveguide, most optical mode field is confined to the LN layer. It is important to note that despite most of the light being confined in the LN layer, the light's polarization can be effectively manipulated by the photonic metamaterials on the Si_3N_4 layer, as shown in the following **Figures 1a–c**.

The Si_3N_4 layer is regarded as a homogeneous medium with a polarization-insensitive refractive index, and according to Rytov's formula,^[28] the equivalent refractive indices of photonic metamaterial on the Si_3N_4 layer are different along different metamaterial directions, which can be written as:

$$n_{//}^2 = D \cdot n_{\text{SiN}}^2 + (1 - D) \cdot n_{\text{Air}}^2 \quad (1)$$

$$1/n_{\perp}^2 = D \cdot 1/n_{\text{SiN}}^2 + (1 - D) \cdot 1/n_{\text{Air}}^2 \quad (2)$$

n_{SiN} and n_{Air} represent the material refractive indices of Si_3N_4 and air, respectively, D represents the duty cycle, $n_{//}$ and $n_{\perp}$ represent the equivalent refractive indices of the metamaterial along with the parallel and perpendicular electric field polarizations to the metamaterial periodic interface, respectively. **Figure 1e** shows the equivalent refractive index versus duty cycle of the photonic metamaterial relation curve calculated from the above equations. Since TE_0 , TM_0 light vibrates in different directions. For TM_0 mode with the electric field component along the X-direction axis, the refractive index is $n_{//}$ for both the metamaterials arranged periodically along the Y-direction and the metamaterials arranged periodically along the Z-direction. For TE_0 mode with the electric field component along the Z-direction axis, the refractive index of the metamaterial arranged periodically along the Y-direction is $n_{//}$ and the refractive index of the metamaterial arranged periodically along the Z-direction is $n_{\perp}$, as shown in **Figure 1f**.

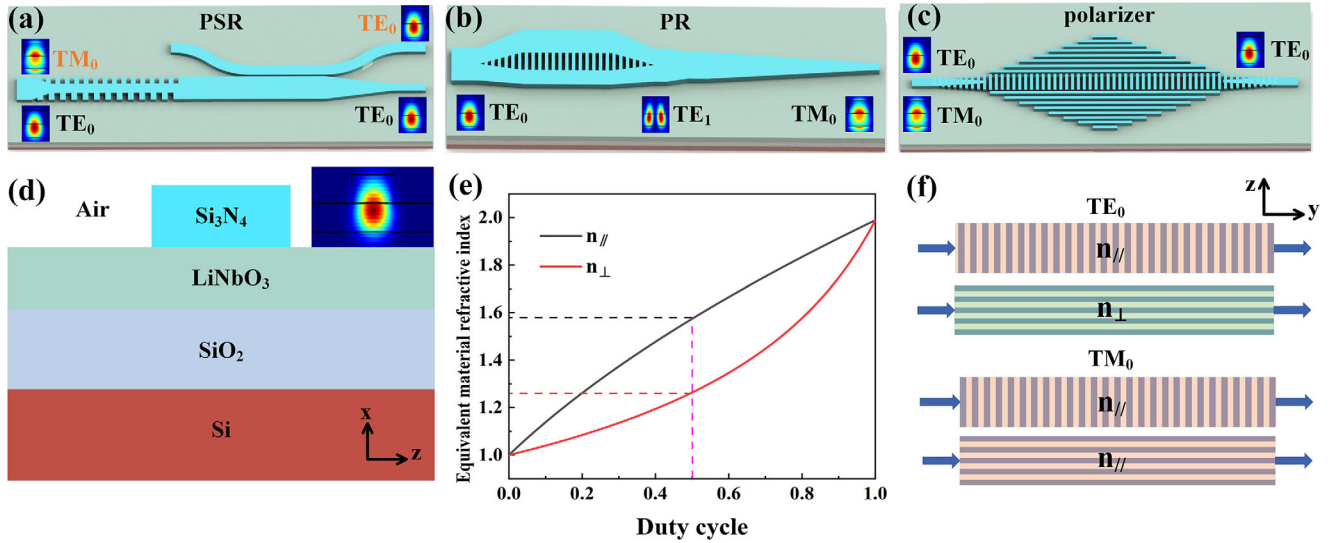


Figure 1. a–c) The schematic of the proposed PSR, PR, and polarizer. d) Cross-section of the hybrid waveguide along the Y-direction. e) The equivalent refractive index of the photonic metamaterial waveguide (PMW) varies with the duty cycle of the PMW. f) Equivalent refractive indices of TE₀ and TM₀ polarization modes in metamaterials with different alignment directions.

For the simulation and design of the photonic components, we use the mode expansion procedure based on the eigenmodes, which can be used to calculate the optical distributions in an arbitrary waveguide such as single and multimode waveguides.^[29] The optical field distribution in a waveguide can be written as a superposition of different eigenmodes, and each eigenmode can be written in the form of its amplitude and phase. For a silicon nitride and LN hybrid waveguide propagating along the Y-direction, the propagating field can be described in Equation (3)

$$E(x, y, z) = \sum_p A_p(y) \psi_p(x, z) e^{-i\beta_p y} \quad (3)$$

$A_p(y)$ represents the complex transmission coefficient of the p-mode along the propagation direction, and β_p and $\psi_p(x, z)$ represent the propagation constant and the electric field distribution, respectively. In the process of polarization rotation and mode conversion, the proportion of different modes of light in the multimode waveguide will change. According to the classical coupled mode theory,^[30,31] the dielectric perturbation generated by photonic metamaterials can be expressed by the coupling coefficient κ

$$\kappa_{pq}(y) = \frac{\omega}{4} \iint \psi_p^*(x, z) \Delta\epsilon(x, y, z) \psi_q(x, z) dx dz \quad (4)$$

$\Delta\epsilon(x, y, z)$ represents the dielectric perturbation

$$\Delta\epsilon(x, y, z) = \begin{cases} \epsilon_0 (n_{\text{Air}}^2 - n_{\text{SiN}}^2); & \text{dielectric material} \\ 0; & \text{none} \end{cases} \quad (5)$$

ϵ_0 represents the free-space dielectric constant. n_{Air} and n_{SiN} represent the material refractive indices of air and silicon nitride,

respectively. The complex transmission coefficient $A_p(y)$ can be obtained from the coupled mode theory

$$\frac{dA_p(y)}{dy} = -i \sum_p \kappa_{pq}(y) A_p(y) e^{-i(\beta_q - \beta_p)y} \quad (6)$$

Thus, the complex transmission coefficients of each mode can be obtained. The energy proportion of each mode can be calculated using $|A_p(y)|^2$, and thus the polarization mode conversion efficiency of the photonic metamaterials can be calculated accurately.

2.2. Polarization Splitter Rotator

Lithium niobate is a typical anisotropic material. For the X-cut wafer, the propagation direction of light is along with the Y- or Z-direction, and the mode hybridization characterizations are also different along different propagation directions. As a proof of concept, we design the PSR along the Y-direction in this work, and its working principle is as follows: the TM₀ light injected at the input port is first rotated to TE₁ after passing through the photonic metamaterial zone, then converted to TE₀ by asymmetric directional coupler (ADC), and finally output at the cross port. The TE₀ light injected at the input port can go through the proposed PSR to the output port without disturbing it. Here, the chirped subwavelength gratings (CSWGs) along two sides of the bus waveguide are designed as photonic metamaterial structures to achieve TM₀-to-TE₁ polarization rotation (as shown in Figure 2a (i)), where the overlap integral of the TM₀ and TE₁ modes is larger than that's in the traditional waveguide. The nonuniform refractive index distribution along the Z-direction is formed and the mode dispersion curves are becoming smoother due to the introduction of the CSWGs. In this way, the coupling length of the PSR can be reduced, and its operating bandwidth can be expanded.

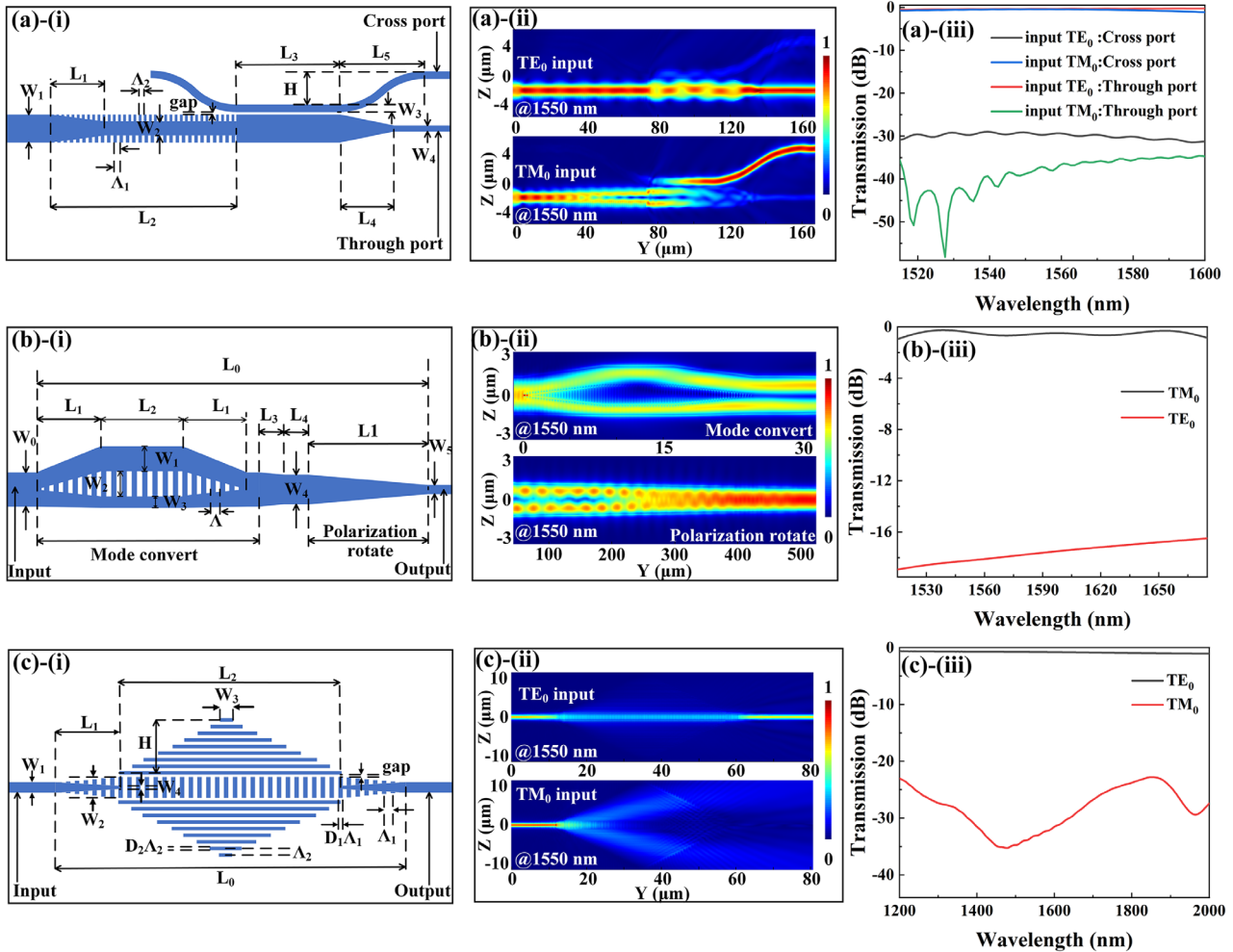


Figure 2. The top views with marked structural parameters and simulated results of the proposed PSR, PR, and polarizer. a (i), b (i), c (i)) top views of the proposed PSR, PR, and polarizer, respectively, a (ii), b (ii), c (ii)) simulated optical field distributions at 1550 nm of the proposed PSR, PR, and polarizer, respectively, and a (iii), b (iii), c (iii)) simulated response spectra of the proposed PSR, PR, and polarizer, respectively.

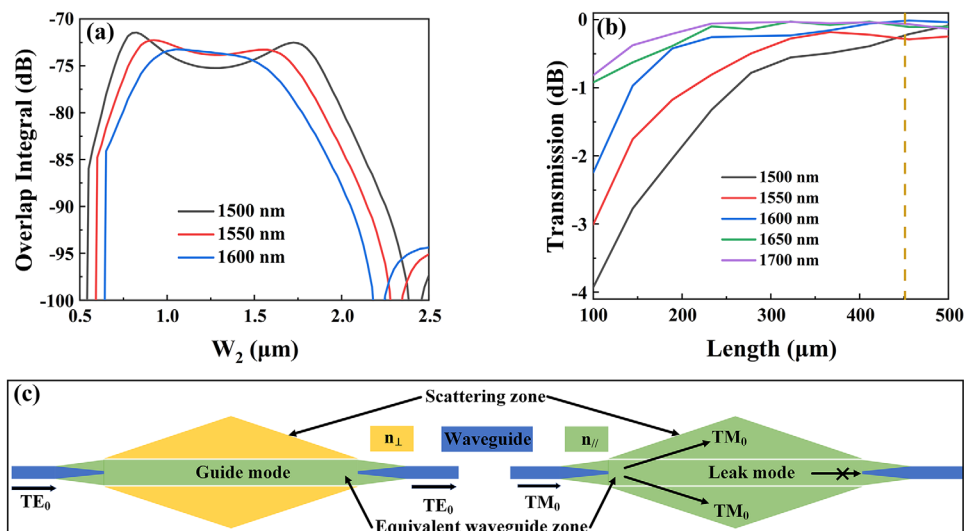


Figure 3. a) The overlap integral as a function of bus waveguide widths at various wavelengths. b) The transformation efficiency as a function of adiabatic cone length at various wavelengths. c) Structure with different equivalent refractive indices for the TE_0 and TM_0 modes, respectively.

Table 2. Physical structure parameters of the polarization splitter rotator.

Physical structure parameters of the polarization splitter rotator				
$W_1 = 2.84 \mu\text{m}$	$W_2 = 1.42 \mu\text{m}$	$W_3 = 1.2 \mu\text{m}$	$W_4 = 0.6 \mu\text{m}$	$H = 5 \mu\text{m}$
$L_1 = 5 \mu\text{m}$	$L_2 = 78 \mu\text{m}$	$L_3 = 28 \mu\text{m}$	$L_4 = 15 \mu\text{m}$	$L_5 = 50 \mu\text{m}$
$\Lambda_1 = 400 \text{ nm}$	$\Lambda_2 = 300 \text{ nm}$	gap = 200 nm	$D = 0.55$	

The bus waveguide width W_1 of the proposed PSR is designed as $2.84 \mu\text{m}$. The periods of two CSWGs along two sides of the bus waveguide are designed to be 0.3 and $0.4 \mu\text{m}$, respectively, and their duty cycle is designed to be the same as 0.55 . As shown in **Figure 3a**, we can see that the overlap integral of the TM_0 and TE_1 is closely related to the width W_2 of the bridge waveguide integrated in the middle of two CSWGs. Theoretically, the greater the overlap integral, the higher the coupling efficiency, and the shorter the length of the bridge waveguide is. However, the width of the bridge waveguide also affects the transmission loss and operation bandwidth of the proposed device. Therefore, considering the tradeoff among transmission loss, operation bandwidth, and footprint size et al, the optimized physical structure parameters of the proposed PSR are shown in **Table 2**. (The detailed analysis can be seen in Section S1, Supporting Information)

To reduce the crosstalk induced by the remaining TM_0 mode signal in the bus waveguide, a TE-pass polarizer formed by a straight waveguide with the width of $W_4 = 0.6 \mu\text{m}$ is connected to the bus waveguide by an adiabatic taper. The polarizer is formed by a straight waveguide of width, where the TE_0 mode signal can pass through it with low loss, while most of the TM_0 mode signal will leak laterally. According to the simulation results, the transmission loss of the TM_0 mode at a wavelength of 1550 nm is estimated to be over 100 dB cm^{-1} due to the lateral leakage, and the transmission loss of the TE_0 mode is negligible.^[32]

The 3D finite-difference time-domain (3D-FDTD) method^[33] is employed to simulate the performance of the proposed PSR. The optical field distributions of the PSR under different polarization modes are shown in **Figure 2a** (ii), from which we can see the proposed device indeed implements the function of the PSR. The simulated response spectra of the PSR as shown in **Figure 2a** (iii) show a low insertion loss (IL) of $< 0.6 \text{ dB}$ and a high extinction ratio (ER) $> 28 \text{ dB}$ for the TE_0 mode from the wavelength of 1515 to 1600 nm , and for the TM_0 mode, the IL is below 1 dB , and the ER is better than 33 dB . The TE-polarized light in the two output ports exhibits a mode purity of over 99% in simulation.

2.3. Polarization Rotator

In this work, we also propose and demonstrate a PR using SWG-based photonic metamaterial, which includes two key parts: mode conversion and polarization rotation zones along the Y-direction of the X-cut LNOI wafer, as shown in **Figure 2b** (i). The injected TE_0 mode at the input port is first converted to TE_1 through the mode conversion zone and then rotated to TM_0 by the polarization rotation zone of the proposed PR.

Two SWG-assisted asymmetric Y-branches are cascaded by two SWG-assisted waveguides with different widths to form the mode conversion zone. Generally, the mode field mismatch occurs at the junction of the Y-branch, which will increase the

transmission loss and decrease the operation bandwidth of the proposed device. To overcome this issue, the smaller angle of the Y-branch is more desirable for forming an adiabatic transition zone, while this will induce a great challenge for the fabrication process. Here, the metamaterial-assisted Y-branches are employed to construct the proposed PR to increase evanescent coupling efficiency, reduce phase adaptation at the interface, radiation loss, and unbalance of the splitting optical power. More importantly, it becomes easier to achieve a more compact size of the proposed PR due to the introduction of photonic metamaterial.

The input TE_0 mode is first split into two beams with equal power by the first asymmetric Y-branch, after that they are combined into a beam by the second asymmetric Y-branch after passing through two asymmetric waveguides. Due to the asymmetric Y-branches, the two beams of light will have phase differences $\Delta\varphi_1$ and $\Delta\varphi_3$ when they pass through the two cascaded Y-branches, respectively, as shown in **Figure 2b** (i). The total phase difference between the two beams can be expressed as

$$\Delta\varphi = \Delta\varphi_1 + \Delta\varphi_2 + \Delta\varphi_3 = \pi \quad (7)$$

$\Delta\varphi_2$ represents the phase difference introduced through two different waveguides. When the total phase difference between the two TE_0 beams is π , they can be combined in one waveguide in the form of TE_1 by the second asymmetric Y-branch. Since we use the same two asymmetric Y-branch in the proposed device, the total phase difference induced by two Y-branches is zero due to their reverse connection, which can be expressed as

$$\Delta\varphi_1 + \Delta\varphi_3 = 0 \quad (8)$$

$\Delta\varphi_2$ is the phase difference introduced by two waveguides of different widths and the same length. The effective refractive index of the same mode is different in different widths of waveguides. Two waveguides with different effective refractive indices produce a phase difference^[34]

$$\Delta\varphi_2 = \frac{2\pi (n_{\text{eff}1} - n_{\text{eff}2}) L_2}{\lambda} \quad (9)$$

$n_{\text{eff}1}$ and $n_{\text{eff}2}$ represent the effective refractive index of the TE_0 mode in the upper wide waveguide and the lower narrow waveguide, respectively, as shown in **Figure 2b** (i). According to Equation (9), we can design the length of L_2 to achieve $\Delta\varphi = \pi$.

The mode hybridization phenomenon caused by the vertical asymmetry of the Si_3N_4 -LNOI waveguide is utilized to achieve polarization rotation,^[35] and mode hybridization will occur when the waveguide width is $\approx 2.1 \mu\text{m}$, where an indistinguishable hybridization of two polarizations exists in the waveguide. The conversion between TM_0 mode and TE_1 mode can be realized using mode hybridization. In this work, mode hybridization is realized using an adiabatic taper whose width changes from 2.45 to $1.75 \mu\text{m}$ with a length of $450 \mu\text{m}$ (**Table 3** and **4**). (The details can be seen in Section S2, Supporting Information)

Figure 2b (ii) shows the simulated power distributions of the PR at 1550 nm . The TE_0 mode is first converted into the TE_1 mode and adiabatically converted to the TM_0 mode. **Figure 2b** (iii) shows the simulated response spectra of this device. This de-

Table 3. Physical structure parameters of the polarization rotator.

Physical structure parameters of the polarization rotator				
$W_0 = 2.84 \mu\text{m}$	$W_1 = 1.5 \mu\text{m}$	$W_2 = 1.5 \mu\text{m}$	$W_3 = 0.8 \mu\text{m}$	$W_4 = 2.45 \mu\text{m}$
$W_5 = 1.75 \mu\text{m}$	$L_0 = 515 \mu\text{m}$	$L_1 = 8.4 \mu\text{m}$	$L_2 = 7.65 \mu\text{m}$	$L_3 = 20 \mu\text{m}$
$L_4 = 20 \mu\text{m}$	$L_5 = 450 \mu\text{m}$	$\Lambda = 300 \text{ nm}$	$D = 0.5$	

Table 4. Physical structure parameters of the polarizer.

Physical structure parameters of the polarizer				
$L_0 = 70 \mu\text{m}$	$L_1 = 12.5 \mu\text{m}$	$L_2 = 45 \mu\text{m}$	$H = 10 \mu\text{m}$	gap = 160 nm
$W_1 = 1 \mu\text{m}$	$W_2 = 1.8 \mu\text{m}$	$W_3 = 1 \mu\text{m}$	$W_4 = 0.35 \mu\text{m}$	
$\Lambda_1 = 300 \text{ nm}$	$\Lambda_2 = 320 \text{ nm}$	$D_1 = 0.5$	$D_2 = 0.5$	

vice shows a low IL of < 1.1 dB and high ER > 15.5 dB for the TE₀ mode to TM₀ mode covering the wavelength range of 1515 to 1675 nm.

2.4. Polarizer

In this work, we propose a TE-pass polarizer along the Y-direction using photonic metamaterials arranged in different directions, and the photonic metamaterials arranged in the transverse direction are named as the equivalent waveguide zone, the photonic metamaterials arranged in the vertical direction are named as the scattering zone, which is shown as Figure 3c. Light in different polarization states will exhibit different transmission losses induced by photonic metamaterials consisting of gratings with different directions of periodic alignment. Based on this principle, a TE-pass polarizer can be designed, as shown in Figure 3c. For the TE₀ mode, the equivalent refractive index of its equivalent waveguide zone is different from that of the scattering zones. The refractive index of the equivalent waveguide zone is larger than that of both sides, so the light will be confined to the equivalent waveguide zone and not leaked to both sides, passing through the photonic metamaterial zone with low loss. For the TM₀ mode, the refractive indices of the equivalent waveguide zone and the scattering zones are the same. When TM₀ mode is input into the proposed device, it will leak to the scattering zones on both sides,

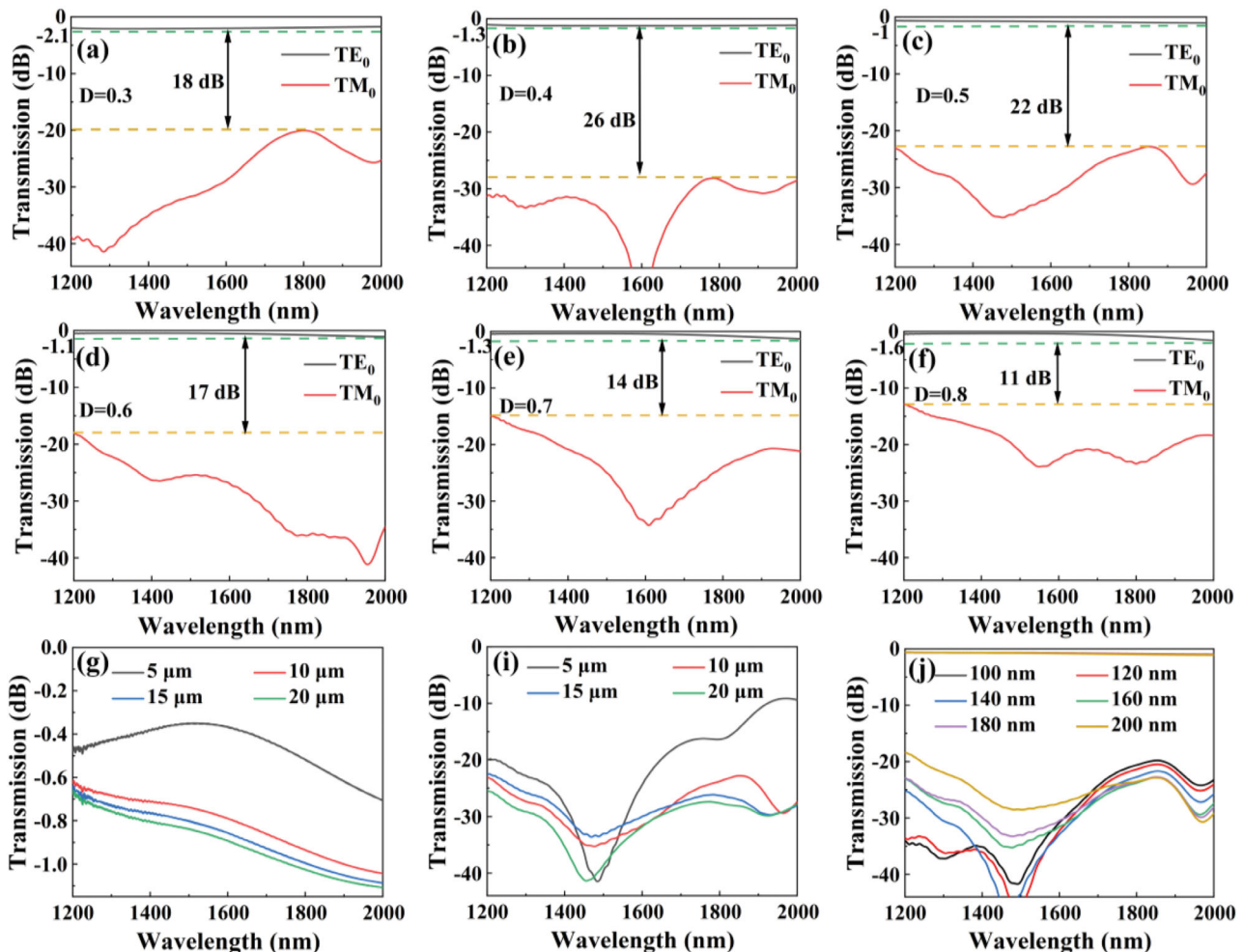


Figure 4. a–f) Simulated response spectra of the proposed polarizer under different duty cycles. g, i) Simulated response spectra of the proposed polarizer with different heights at TE₀ input and TM₀ input, respectively. j) Simulated response spectra of the proposed polarizer with different gaps at TE₀ input (Six curves above) and TM₀ input (Six curves below), respectively.

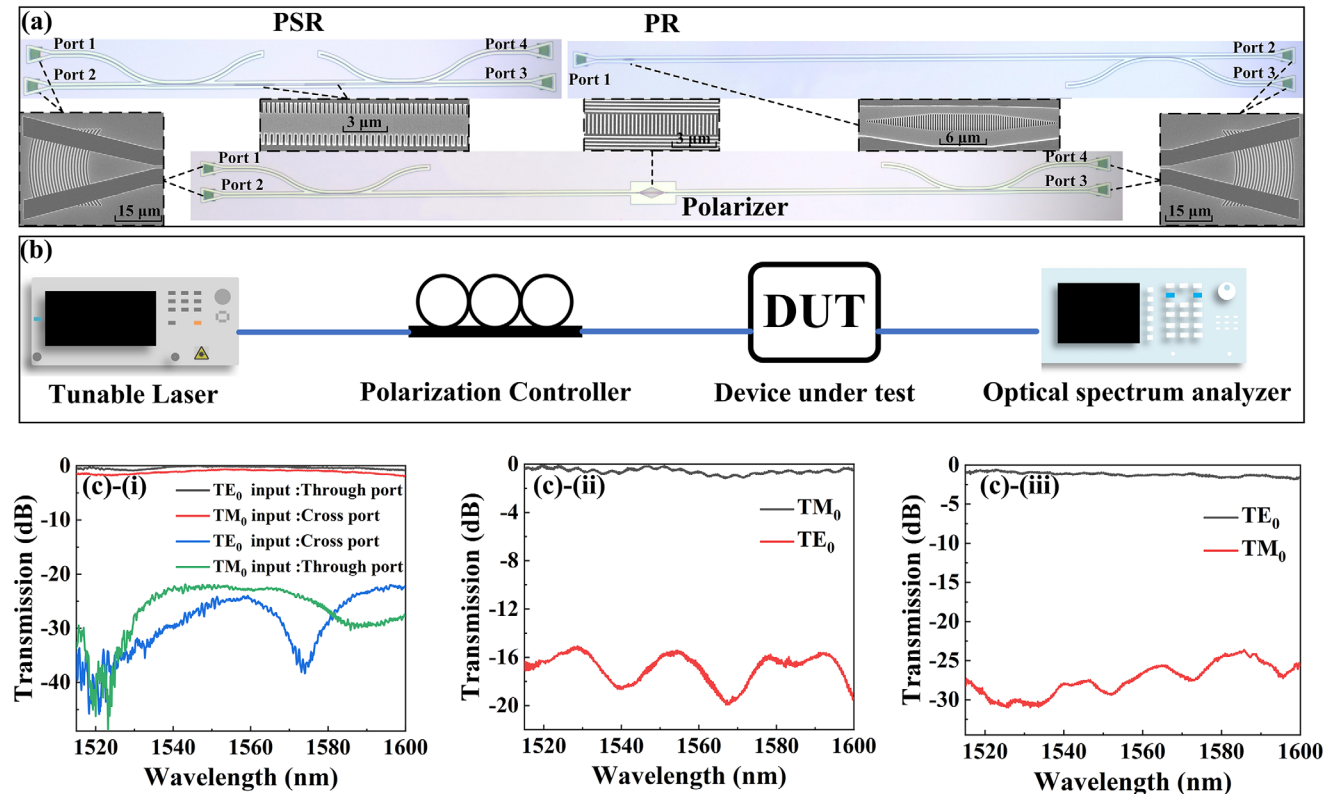


Figure 5. a) Optical microscope images and SEM images of the fabricated devices. b) Experimental setup for characterizing the proposed devices. c) Measured response spectra for the fabricated i) PSR, ii) PR, and iii) Polarizer.

and it cannot propagate along the original direction. Therefore, the TE₀ mode can pass through the photonic metamaterial with low transmission loss in the form of the guided mode, while the TM₀ mode will behave as a leakage mode and leak to both sides.

From Figure 1e, we can see the equivalent refractive index contrast between the $n_{||}$ and $n_{\perp}$ for both TE₀ and TM₀ polarizations is dependent on the duty cycle of the designed subwavelength gratings. To optimize the performance of the proposed device, the physical structure parameters such as duty cycle, the height of the scattering zone H , and the gap are defined by the 3D-FDTD simulation method, and the simulation results are shown in Figures 4a–j.

Figure 2c (ii) shows the normalized power distributions of the TM₀ and TE₀ modes, respectively. The input TM₀ becomes leaky mode, while the input TE₀ keeps guided mode due to the introduction of the photonic metamaterial along different directions. The response spectra of the proposed device for TE₀ and TM₀ modes are shown in Figure 2c (iii), which shows the device has a low IL of < 1.05 dB and a high ER of > 21.5 dB covering the wavelength range of 1200–2000 nm.

3. Fabrication and Characterization

The proposed devices are fabricated on the commercial X-cut LNOI wafer with the LN thin film thickness of 300 nm provided by the “NANOLN.” A silicon nitride thin film layer with a thickness of 300 nm is first deposited on the top of the LNOI wafer using reactive sputtering. Then the proposed devices are patterned

on the silicon nitride thin film layer by electron beam lithography (EBL) and inductively coupled plasma etching (ICP) processes. High-performance grating couplers^[36] are integrated at the terminals of the proposed devices to achieve the coupling between the standard single-mode fiber (core diameter of 9 μm and cladding diameter of 125 μm) and the input/output ports of the proposed. The zoom-in scanning electron microscope (SEM) images of the devices and the grating coupler are shown in Figure 5a, from which we can see that the micro-structures of the grating coupler and the polarization-manipulating devices are fabricated successfully on the same chip based on the mature fabrication process of silicon nitride.

A tunable laser (TL), a polarization controller (PC), and an optical spectrum analyzer (OSA) are employed to characterize the performance of the fabricated device, which is shown in Figure 5b. In the experiment, the laser light is first transmitted into the PC and then coupled into and out of the chip via the input and output grating couplers, finally the transmission spectra are recorded by the OSA. The measured results are then normalized by using the transmission of the reference straight waveguide. The experimental results of the proposed PSR, PR, and polarizer are shown in Figure 5c, from which we can see that the PSR shows IL < 1 dB and ER > 20 dB for TE₀ mode and IL of < 1.6 dB and ER > 20 dB for TM₀ mode in the wavelength range of 1515 to 1600 nm. The proposed PR shows IL of < 1 dB and ER > 14 dB in the wavelength range of 1515 to 1600 nm. The proposed polarizer shows IL of < 1.75 dB and ER > 20 dB in the wavelength range of 1515 to 1600 nm.

Table 5. Performance comparison of polarization manipulating devices on LNOI.

Function	Structure	Length [μm]	ER [dB]	IL [dB]	Bandwidth [nm]	Exp or Sim
PSR	Adiabatic taper	431	> 20	< 0.65	80	Exp
	+ ADC ^[37–39]	8000	≈8 TE ≈10 TM	≈2 TE ≈3 TM	130 (1275–1355)	Exp
		240	> 26.6 TE > 10.6 TM	0.91 TE (1550) 0.38 TM (1550)	50 (1520–1570)	Exp
	Adiabatic taper	440	> 26.6 TE > 19.6 TM	< 1	60 (1520–1580)	Exp
	+ Asymmetric Y-branch ^[40]					
	Adiabatic taper	679	> 20 TE > 10 TM	< 1.5	47 (1502–1649)	Exp
	+ Y-branch ^[41]					
	Adiabatic taper	> 1200	> 20	< 1.5	35 (1530–1565)	Exp
	+ Adiabatic coupler ^[42]					
	Mode evolution taper + ADC ^[43]	405	> 22 TE > 11 TM	< 2 TM < 1 TE	160 (1480–1640)	Exp
PR	Mode evolution waveguide + ADC ^[35]	620	> 17.8 TE > 10.6 TM	< 0.85 TE < 1.49 TM	40 (1525–1565)	Exp
	This work	160	> 20	< 0.85 TE 0.14 (1550) TE < 1.83 TM 0.73 (1550) TM	85 (1515–1600)	Exp
	Sb ₂ Se ₃ layer deposited ^[44]	17.7	> 14	< 0.38	107	Sim
	Semi-infinite cover Si ₃ N ₄ layer ^[45]	15.77	> 20	< 0.3	120	Sim
	HPW ^[46]	13.7	> 30	< 1.44	30	Sim
	This work	500	> 15 (Sim) > 14 (Exp)	< 0.82 (Sim) < 1.07 (Exp)	160 (Sim) 85 (Exp)	Exp
TE-pass polarizer	ITO and Si ₃ N ₄ assisted ^[47]	7.5	> 20	< 0.95	125	Sim
	Ge ₂ Sb ₂ Te ₅ assisted ^[48]	7.5	> 25	< 0.5	200	Sim
	Shallow etched waveguide ^[32,49]	1000	108/mm	–	–	Sim
		1000	28.72/mm	–	–	Sim
	Hybrid plasma grating ^[50]	9	> 15	< 3.4	300	Sim
	This work	70	> 21 (Sim) > 20 (Exp)	< 1.05 (Sim) < 1.75 (Exp)	800 (Sim) 85 (Exp)	Exp

IL, Insertion loss; ER, Extinction ratio; Exp: Experiment; Sim: Simulation; ADC: Asymmetric directional coupler; HPW: Hybrid plasmonic waveguide; ITO: Indium tin oxide

Table 5 shows a performance comparison of the reported polarization manipulating devices on LNOI, which shows the designed photonic metamaterial-assisted polarization manipulating devices have excellent performance, demonstrating the powerful optical operation capability of photonic metamaterials.

4. Conclusion

In this work, we demonstrate that photonic metamaterials have powerful optical manipulation capabilities on the LNOI platform. These metamaterials can control the coupling state of different polarizations, provide asymmetry in the vertical direction, reduce the radiation loss at discontinuous interfaces, and effectively manipulate the equivalent refractive index. Using the most promising load material of the LNOI platform, silicon nitride, and its mature manufacturing process, we design and manufacture polarization processing devices based on photonic metamaterial structure, including PSR, PR, and polarizer. Compared with conventional structures, these devices have larger operating bandwidth, lower insertion loss, and higher extinction ratio. These

photonic metamaterials may hold great potential for the development of various applications in nonlinear optics, spin optics, negative index, and zero index materials, and so on. This work is expected to pave the way for photonic metamaterial devices based on the LNOI platform.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

lithium niobate on insulator (LNOI), photonic metamaterial, polarization manipulating, silicon nitride (Si_3N_4)

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